Machine Learning Assignment :- 10

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Q. 10. Assignment on Regression technique.

Download temperature data from the link below.

DATASET LINK :-https://www.kaggle.com/venky73/temperaturesof-india?select=temperatures.csv

This data consists of temperatures of INDIA averaging the temperatures of all places month wise. Temperatures values are recorded in CELSIUS

1. Apply Linear Regression using a suitable library function and predict the Month-wise temperature.

2. Assess the performance of regression models using MSE, MAE and R-Square

3. Metrics

4. Visualize a simple regression model.

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1. Apply Linear Regression using a suitable library function and predict the Month-wise temperature.

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error, mean_absolute_error, r2_score
```

```
df = pd.read_csv("temperatures.csv")
df.head()
```

	YEAR	JAN	FEB	MAR	APR	MAY	JUN	JUL	AUG	SEP	OCT	NOV	DEC	ANNUAL	JAN-FEB	MAR-MAY	JUN-SEP	OCT-DEC
0	1901	22.40	24.14	29.07	31.91	33.41	33.18	31.21	30.39	30.47	29.97	27.31	24.49	28.96	23.27	31.46	31.27	27.25
1	1902	24.93	26.58	29.77	31.78	33.73	32.91	30.92	30.73	29.80	29.12	26.31	24.04	29.22	25.75	31.76	31.09	26.49
2	1903	23.44	25.03	27.83	31.39	32.91	33.00	31.34	29.98	29.85	29.04	26.08	23.65	28.47	24.24	30.71	30.92	26.26
3	1904	22.50	24.73	28.21	32.02	32.64	32.07	30.36	30.09	30.04	29.20	26.36	23.63	28.49	23.62	30.95	30.66	26.40
4	1905	22.00	22.83	26.68	30.01	33.32	33.25	31.44	30.68	30.12	30.67	27.52	23.82	28.30	22.25	30.00	31.33	26.57

Next steps: Generate code with df New interactive sheet

```
months = ["JAN","FEB","MAR","APR","MAY","JUN","JUL","AUG","SEP","OCT","NOV","DEC"]
data = []

for _, row in df.iterrows():
    for i, m in enumerate(months):
        data.append([i+1, row[m]])

long_df = pd.DataFrame(data, columns=["Month", "Temperature"])

X = long_df[["Month"]]
y = long_df["Temperature"]

long_df.head()
```

	Month	Temperature
0	1	22.40
1	2	24.14
2	3	29.07
3	4	31.91
4	5	33.41

Next steps: Generate code with long_df New interactive sheet

2. Assess the performance of regression models using MSE, MAE and R-Square

```
model = LinearRegression()
model.fit(X, y)

y_pred = model.predict(X)

mse = mean_squared_error(y, y_pred)
mae = mean_absolute_error(y, y_pred)
r2 = r2_score(y, y_pred)
```

```
print("Mean Squared Error (MSE):", mse)
```

```
Mean Squared Error (MSE): 10.138413546423768
```

```
print("Mean Absolute Error (MAE):", mae)
```

```
Mean Absolute Error (MAE): 2.6878642126576615
```

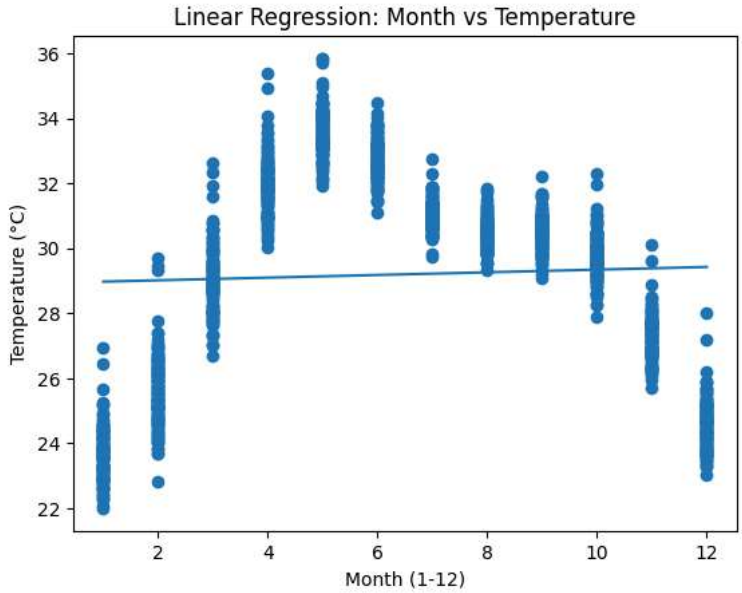
```
print("R-Square (R2 Score):", r2)
```

```
R-Square (R2 Score): 0.0019717978743754028
```

3. Metrics

```
print("Mean Squared Error (MSE):", mse)
print("Mean Absolute Error (MAE):", mae)
print("R-Square (R2 Score):", r2)
```

```
Mean Squared Error (MSE): 10.138413546423768
Mean Absolute Error (MAE): 2.6878642126576615
R-Square (R2 Score): 0.0019717978743754028
```

4. Visualize a simple regression model.

```
# Scatter plot
plt.figure()
plt.scatter(X, y)
plt.plot(X, y_pred)
plt.xlabel("Month (1-12)")
plt.ylabel("Temperature (°C)")
plt.title("Linear Regression: Month vs Temperature")

plt.show()
```